

The Bapat–Sunder Permanent-Minor Matrix on Regular Real Projective Polygons

Zijian Zeng^{1,2}, Houde Liu³, Kuru Ratnavelu^{2,4}, Ong Seng Huat⁵, and Yonghua Xiong^{*1}

¹School of Artificial Intelligence and Automation, China University of Geosciences,
Wuhan 430074, P.R. China

²Institute of Computer Science and Digital Innovation, UCSI University, Kuala
Lumpur 56000, Malaysia

³Tsinghua Shenzhen International Graduate School, Shenzhen 518055, China

⁴Institute of Mathematical Sciences, University of Malaya, Kuala Lumpur 50603,
Malaysia

⁵Sciences, University of Malaya, 50603 Kuala Lumpur, Malaysia

875100501@qq.com; 1002266693@ucsiuniversity.edu.my; liu.hd@sz.tsinghua.edu.cn;
Kurunathan@ucsiuniversity.edu.my; ongsh@um.edu.my

**Corresponding Author: Yonghua Xiong, xiongyh@cug.edu.cn*

Abstract

For a real positive semidefinite matrix $A = (a_{ij})$, let $F_A = (a_{ij} \text{ per } A(i, j))$, where $A(i, j)$ is obtained by deleting row i and column j . The assertion that $\text{per } A$ is the largest eigenvalue of F_A remains a natural real form of a conjecture of Bapat and Sunder. We prove it, with the complete spectrum, for a non-entrywise-nonnegative family in every order. Namely, for

$$A_n = (\cos(\pi(i - j)/n))_{0 \leq i, j < n}, \quad n \geq 3,$$

we show

$$\text{per } A_n = \frac{n!}{2^{n-1}}, \quad \lambda_s(F_{A_n}) = \frac{n!}{2^n} \left\{ \binom{n-1}{s}^{-1} + \binom{n-1}{s-1 \pmod{n}}^{-1} \right\}.$$

Hence the constant mode is uniquely maximal, and the largest nonconstant eigenvalue is exactly $n/[2(n-1)]$ times the permanent. The proof combines a symmetric-tensor permanent identity with deleted-root elementary symmetric functions and a discrete Fourier diagonalization. An independent exact checker reconstructs the permanent and all required minors in cyclotomic quotient rings. We also report a clearly separated numerical reconnaissance over real Gram ranks two through five and orders through fifteen; it produced no counterexample and is not used as proof.

1 Introduction

For an $n \times n$ matrix $A = (a_{ij})$, its permanent is

$$\text{per } A = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

If A is positive semidefinite and $A(i, j)$ denotes the submatrix obtained by deleting row i and column j , Bapat and Sunder considered the matrix

$$F_A = (a_{ij} \text{ per } A(i, j))_{i,j=1}^n. \quad (1)$$

It is positive semidefinite, and the Laplace expansion gives $F_A \mathbf{1} = (\text{per } A) \mathbf{1}$. They asked whether $\text{per } A$ must be its largest eigenvalue [1].

For complex Hermitian matrices the answer is no: Drury gave a rank-two counterexample of order eight and reported that no order-seven example was found [3]. The same paper explicitly isolated the real positive-semidefinite version as a conjecture. It is important not to conflate that question with two nearby statements. Drury's real rank-three example of order sixteen concerns a stronger Schur-power/permanent- Oppenheim statement [2]. More recently, Pioge et al. produced an order-sixteen *complex Hermitian* Gram matrix whose F_A has a violating real Rayleigh vector, and reported an unsuccessful order-fifteen numerical search [4]. Neither construction is a real symmetric F_A counterexample.

Our result is a rigorously solvable infinite stratum of the real question. The matrices A_n below have rank two and are generated by the regular n -gon in the real projective line. Their signs are genuinely frustrated: no diagonal sign switching makes all entries nonnegative. Nevertheless, the entire spectrum of F_{A_n} admits a simple closed form.

2 A symmetric-tensor permanent identity

For $\theta \in \mathbb{R}$, write $x_\theta = (\cos \theta, \sin \theta)^T$ and $q_\theta = e^{2i\theta}$. We use e_k for the k th elementary symmetric function.

Lemma 1. *Let $\alpha_1, \dots, \alpha_m$ and $\beta_1, \dots, \beta_m$ be real. If $G_{pq} = x_{\alpha_p}^T x_{\beta_q}$, then*

$$\text{per } G = \frac{m!}{2^m} e^{i(\sum_p \alpha_p - \sum_q \beta_q)} \sum_{k=0}^m \frac{e_k(q_{\alpha_1}, \dots, q_{\alpha_m}) e_k(q_{\beta_1}, \dots, q_{\beta_m})}{\binom{m}{k}}. \quad (2)$$

Proof. Complexify $\mathbb{R}^2$ and use its orthonormal circular basis. The coefficient of the normalized vector in $\text{Sym}^m(\mathbb{C}^2)$ having k copies of the second circular basis vector is

$$2^{-m/2} e^{-i \sum_p \alpha_p} \frac{e_k(q_{\alpha_1}, \dots, q_{\alpha_m})}{\sqrt{\binom{m}{k}}}.$$

The inner product of two symmetrized product tensors is the permanent of their cross-Gram matrix divided by $m!$. Taking the inner product of these coordinate expansions proves (2). $\square$

3 The exact spectrum

Theorem 2. *Let $n \geq 3$, set $\theta_j = \pi j/n$ and $v_j = x_{\theta_j}$ for $0 \leq j < n$, and put*

$$A_n = (v_i^T v_j)_{0 \leq i, j < n} = (\cos(\pi(i-j)/n))_{0 \leq i, j < n}.$$

Then A_n is a real correlation matrix of rank two,

$$\text{per } A_n = \frac{n!}{2^{n-1}}, \quad (3)$$

and the eigenvalues of F_{A_n} are

$$\lambda_s = \frac{n!}{2^n} \left(\binom{n-1}{s}^{-1} + \binom{n-1}{s-1 \pmod{n}}^{-1} \right), \quad 0 \leq s < n. \quad (4)$$

In particular,

$$\lambda_0 = \text{per } A_n, \text{quad} \max_{1 \leq s < n} \lambda_s = \frac{n}{2(n-1)} \text{per } A_n < \text{per } A_n.$$

Proof. Let $q_j = e^{2i\theta_j} = e^{2\pi i j/n}$. These are all n th roots of unity, so

$$e_k(q_0, \dots, q_{n-1}) = 0 \quad (1 \leq k < n), \quad |e_0| = |e_n| = 1.$$

Lemma 1, applied to the same full list on both sides, gives (3).

Put $m = n - 1$. Removing q_i from the full root set and using

$$0 = e_k(q_0, \dots, q_{n-1}) = e_k(q_0, \dots, \widehat{q_i}, \dots, q_{n-1}) + q_i e_{k-1}(q_0, \dots, \widehat{q_i}, \dots, q_{n-1})$$

shows recursively that

$$e_k(q_0, \dots, \widehat{q_i}, \dots, q_{n-1}) = (-q_i)^k, \quad 0 \leq k \leq m. \quad (5)$$

For $d = j - i$ modulo n and $\delta = \pi d/n$, substitution of (5) into Lemma 1 yields

$$\text{per } A_n(i, j) = \frac{m!}{2^m} \sum_{k=0}^m \frac{\cos((2k+1)\delta)}{\binom{m}{k}}. \quad (6)$$

The expression before taking real parts is already real: pairing k with $m - k$ uses $m = n - 1$ and $e^{2in\delta} = 1$.

Set $c_k = \binom{m}{k}^{-1}$. Multiplying (6) by $a_{ij} = \cos \delta$ and applying the product-to-sum identity gives

$$(F_{A_n})_{ij} = \frac{m!}{2^{m+1}} \sum_{k=0}^m c_k \left\{ \cos\left(\frac{2\pi kd}{n}\right) + \cos\left(\frac{2\pi(k+1)d}{n}\right) \right\}. \quad (7)$$

Thus F_{A_n} is circulant. Reading subscripts modulo n , the coefficient of Fourier frequency s in (7) is $c_s + c_{s-1}$. The symmetry $c_{n-1-k} = c_k$ then gives

$$\lambda_s = \frac{n(n-1)!}{2^n} (c_s + c_{s-1}),$$

which is (4).

For $s = 0$, $c_0 + c_{n-1} = 2$. For every nonconstant mode, the two summands cannot both equal one. The largest such value occurs at $s = 1, n - 1$ and is $1 + 1/(n - 1)$. Division by the constant-mode coefficient 2 proves the final assertion. $\square$

Remark 3 (Not an entrywise-nonnegative case). *For $n \geq 3$, consecutive edges of the cycle of Gram lines have positive inner product, while the closing edge has negative inner product. The sign product around this cycle is negative and is invariant under diagonal sign switching. Hence no switching makes every nonzero entry of A_n nonnegative.*

4 Exact computational audit

The proof is analytic and valid for every n . We additionally provide a definition-level audit that imports no discovery code. For $3 \leq n \leq 8$ it sets $t = e^{\pi i/n}$ and represents

$$2(A_n)_{ij} = t^{i-j} + t^{j-i}$$

in the exact quotient ring $\mathbb{Q}[t]/(t^n + 1)$. A subset dynamic program computes the full permanent and every deleted-row/deleted-column permanent. The resulting identities are reduced modulo the cyclotomic polynomial Φ_{2n} , and formal Fourier substitution recovers exactly the integer-scaled spectrum in (4). The checker also rejects deliberate wrong-phase, wrong-scale, and wrong-binomial variants. Running under normal Python and under optimization produces byte-identical JSON reports.

This finite audit checks the derivation; it is not used to extrapolate a finite numerical pattern to all n .

5 Numerical reconnaissance and scope

Before the root-of-unity reduction was isolated, we evaluated real unit Gram matrices of orders 6 through 15 and ranks 2 through 5. The evaluator uses the same symmetric-tensor identity but floating-point arithmetic. A random-restart hill-climbing pass made 99,850 objective evaluations and found no strict violation. At order 15, the largest observed nonconstant ratios were as follows.

Gram rank	largest observed nonconstant eigenvalue divided by per A
2	0.99697849
3	0.97266096
4	0.97413850
5	0.94417930

These values are discovery evidence only. They neither exclude a counterexample nor establish a minimum dimension. Theorem 2 is the rigorous outcome: a nontrivial infinite real rank-two family with a complete exact spectrum. The unrestricted real problem, the full rank-two cone, and a rigorous exclusion of any complete order remain open.

6 Reproducibility statement

The source archive contains the theorem proof, the independent cyclotomic verifier, its exact JSON output, fail-closed attacks, the numerical discovery engine and frozen scan summaries, and SHA-256 manifests. No floating-point quantity is used in the proof of Theorem 2.

References

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